

Program 8: Create a Kubernetes pods using YAML dependencies

File structure:

Nodejs-configmap.yaml

Nodejs-pod.yaml

server.js

server.js

```
const http = require("http")
```

```
const port = 3000
```

```
const server = http.createServer((req,res)=>{
```

```
  res.end("Hello from Node.js running")
```

```
});
```

```
server.listen(port,()=>{
```

```
  console.log(`Server is running in port ${port}`);
```

```
})
```

Nodejs-configmap.yaml

```
apiVersion: v1
```

```
kind: ConfigMap
```

```
metadata:
```

```
  name: nodejs-app-config
```

```
data:
```

```
  server.js: |
```

```
    const http = require("http");
```

```
    const port = 3000;
```

```
    const server = http.createServer((req, res) => {
```

```
      res.end("Hello from Node.js running inside kubernetes");
```

```
    });
```

```
    server.listen(port, () => {
```

```
      console.log(`Server is running on port ${port}`);
```

```
    });
```

Nodejs-pod.yaml

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: program-8
```

```
  labels:
```

```
    app: nodejs
```

```
spec:
```

```
  containers:
```

```
    - name: nodejs
```

```
      image: node:18
```

command: ["node", "/usr/src/app/server.js"]

volumeMounts:

- name: app-code

mountPath: /usr/src/app

ports:

- containerPort: 3000

volumes:

- name: app-code

configMap:

name: nodejs-app-config

Start minikube

#minikube start

Check the status to verify all the services are running

#minikube status

Create the pod from YAML files

#kubectl apply -f nodejs-configmap.yaml

#kubectl apply -f nodejs-pod.yaml

List the pods

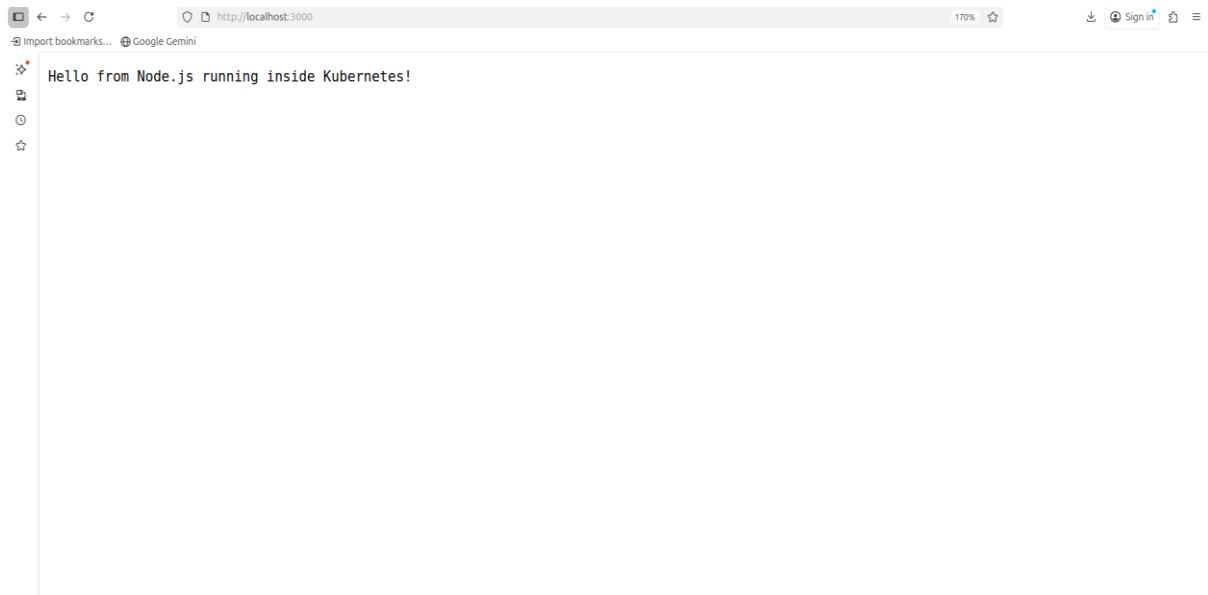
#kubectl get pods

Forward the port to see the nodejs output in browser

#kubectl port-forward pod/program-8 3000:3000

```
1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program8
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program8
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ nano server.js
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ nano nodejs-configmap.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ nano nodejs-pod.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config unchanged
Warning: Detected changes to resource nodejs-pod which is currently being deleted.
pod/nodejs-pod configured
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-9x2qv   1/1     Running   0           44m
nginx-deployment-6f9664446b-bqrs6   1/1     Running   0           44m
nodejs-pod                           0/1     Terminating 0           16m
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl delete pod nodejs-pod --grace-period=0 --force
Warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
pod "nodejs-pod" force deleted from default namespace
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-9x2qv   1/1     Running   0           45m
nginx-deployment-6f9664446b-bqrs6   1/1     Running   0           45m
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config unchanged
pod/nodejs-pod created
1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-9x2qv   1/1     Running   0           45m
nginx-deployment-6f9664446b-bqrs6   1/1     Running   0           45m
nodejs-pod                           0/1     ContainerCreating 0           7s
nodejs-pod                           1/1     Running     0           11s
^C1RV24MC036_gayatri_v_k@ubuntu:~/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::]:3000 -> 3000
Handling connection for 3000
```

Go to browser and search for the url `http://localhost:3000` to see the output



Deleting resources

```
#kubectl delete pods --all
```

```
#kubectl delete svc--all
```

```
#kubectl delete deployments --all
```

